Mariem Kbaier

Computer Science Engineering Student | AI & Data | Tunis, Tunisia

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Professional Summary

Seeking a final-year internship (PFE) in AI & Data. As a 3rd-year student in the computer science engineering program at ENIS, I am interested in designing intelligent solutions, analyzing data and solving real-world problems through emerging technologies.

Education

- **ENIS** – Engineering Degree in Computer Science — Sfax, Tunisia | 2024 – Present
- **IPEIT** – Preparatory Cycle (MP) — Tunis, Tunisia | 2021 – 2024

Experience

- **AgriMent** – Multi-agent AI platform for agricultural decision support — repo
  Talan Tunisia – Summer Camp — *Tunis | July – August 2026*
  - Designed a multi-agent platform of **7 agents** driven by a **supervisor agent** that routes farmer and buyer requests to specialized agents: regulations and subsidies, plot analysis (soil, weather, NDVI, satellite), business scenarios (ROI, risk), crop monitoring, commodity market and waste valorization. **2nd out of 10 teams**.
  - Built a **hybrid RAG** pipeline (Mistral embeddings + BM25 + RRF) over Légifrance, FranceAgriMer and CERFA forms, with sourced answers.
  - Fine-tuned a **TempCNN** crop classifier on Sentinel-2 imagery (**80.3 % accuracy**, versus 78.5 % for the pre-trained model), integrated into the plot analysis to cross-check declared crops.
  - Implemented safeguards: **three-level transparency** (sourced answer, ML prediction, deterministic calculation), **numerical audit** of generated figures and an LLM security filter (prompt injection, banned products, regulatory ceilings), aligned with the EU AI Act.

  LangGraph, Python, FastAPI, Pydantic, Mistral, Hybrid RAG (BM25 + RRF), Qdrant, ChromaDB, PyTorch, TempCNN, Sentinel-2, PostgreSQL/PostGIS, React, Docker
- **Full-Stack Technical Internship**
  BNS Engineering — *Tunis | June – August 2025*
  - Developed a secure **Spring Boot REST API** in a microservices architecture, with a React.js frontend, MySQL and Docker deployment, for a self-service **service catalog**.
  - Integrated a **semantic chatbot** for natural-language search in the catalog and user assistance.

  Spring Boot, REST APIs, React.js, MySQL, Docker, Git, Agile/Scrum

Projects

- **VORTEX** – Multi-agent BDD testing for microservices *(PFA · submission in progress, ICAART 2027)*
  repo microservices
  - Designed a **multi-agent pipeline (LangGraph)** that turns User Stories into executable **Gherkin/Cucumber** test suites for Spring Boot microservices.
  - Integrated a **RAG (ChromaDB)** mechanism over **55,000+ real BDD scenarios**, raising scenario coverage from **20 %→100 %** and pass rate from **32 %→81.2 %**.
  - Implemented an **autonomous self-repair loop** and benchmarked several LLMs.
  - Developed a **Spring Boot microservices backend** secured with JWT and measured code coverage with **JaCoCo**.

  LangGraph, LangChain, Groq API, ChromaDB, Java, Spring Boot, Cucumber, RestAssured, JaCoCo, Maven
- **AI Performance Analysis Agent** – KPI & Power BI — repo
  - Designed a **multi-agent pipeline orchestrated with LangGraph** that reads a Power BI model (.pbix: 10 tables, 9 DAX measures, over **68,000 sales rows**) directly and recomputes monthly KPIs in Pandas using the same formulas as the dashboard.
  - **Anomaly detection** via a 6-month rolling z-score, confirmed by the year-over-year gap, with a configurable threshold and minimum order volume.
  - **Automatic explanation** of each alert: price / volume / mix breakdown of revenue, identification of the category, product and region involved, and share of sales on promotion.

Python, Pandas, LangGraph, pbixray, Power BI, Power Query, DAX

- **Job Matcher** – Multi-agent AI system for CV/job matching *(Talan Tunisia)* repo
  - Developed a candidate–job matching pipeline with evidence-backed explanations, fed with real job postings through the **Apify API** exposed as an **MCP tool**.
  - Implemented **deterministic scoring** (embedding similarity + skill overlap) independent of LLM generation, with a **code-verified citation system** linking each explanation to the source text.

  LangGraph, MCP (FastMCP), Apify API, Groq (Llama 3.3 70B), Sentence-Transformers, ChromaDB, Streamlit
- **Customer Behavior Analysis & Churn Prediction** repo
  - Built an end-to-end ML pipeline on **4,372 customers**: churn prediction, segmentation and spending forecasting.
  - **XGBoost** for churn (**F1: 0.955**), **K-Means/DBSCAN** for segmentation and **Random Forest** for spending.
  - Deployed the models through a **Flask REST API** with an interactive dashboard.

  Python, Pandas, Scikit-Learn, XGBoost, K-Means, DBSCAN, Random Forest, Flask
- **FogWatch** – Attack detection through cross-node correlation in Fog repo
  - **Collaborative IDS** on a 3-site fog architecture: Suricata detects locally, **MQTT** shares observations, and a Python correlator sums attempts per IP to spot "low and slow" SSH attacks.
  - Three components: **fog-agent** (alert normalization), **correlator** (global threshold, anti-spam cooldown) and **response-agent** (nftables blocking with TTL and whitelist).
  - Deployed on **K3s** (Docker), with sites linked through **Tailscale**; real-time MQTT/WebSocket dashboard.

  Suricata, MQTT (Mosquitto), K3s, Docker, Tailscale, nftables, Python, scikit-learn
- **AI-Driven Automated QA Pipeline** repo
  - Designed an **end-to-end QA chain**: requirements → AI-generated test cases (strict **Pydantic** validation) → **Jira/Xray** export with requirements ↔ tests traceability.
  - Developed **supervised AI agents**: test generation with execution-based self-correction and failure triage.
  - Orchestrated the pipeline with **GitHub Actions** and **Docker** (tests on every push): pinned versions, LLM call caching, identical execution locally and in CI.

  Python, Playwright, Selenium, pytest, Docker, GitHub Actions, Jira/Xray

Skills

- **AI & Machine Learning:** LangGraph, LangChain, AI agents, RAG (hybrid search, BM25/RRF), multi-agent systems, MCP, LLM evaluation (RAGAS), Sentence-Transformers, ChromaDB, Qdrant, PyTorch, scikit-learn, XGBoost, computer vision (YOLOv8), OCR
- **Data & BI:** Pandas, SQL, Power BI, DAX, KPI analysis, PostgreSQL/PostGIS, MySQL, NoSQL, Kafka, ETL/ELT
- **MLOps & Cloud:** Docker, GitHub Actions (CI/CD), Git
- **Languages & frameworks:** Python, Java, JavaScript, SQL; FastAPI, Flask, Spring Boot, React.js, Streamlit
- **Concepts:** microservices, BDD/testing, MLOps, Agile/Scrum

Awards & Activities

- **2nd place – Talan Summer Camp 2026:** 2nd out of 10 teams for AgriMent, a generative-AI-based agricultural decision-support platform.
- **1st place – Non-Technical Challenge:** Represented the IEEE PES Chapter – ENIS Student Branch as **Webmaster**.
- **1st place – TEC.Week 3.0:** Tunisian Engineering Congress Week.

Certifications, Involvement & Languages

- **Certifications:** Agentic AI Certified Foundations Associate (Oracle) | LangGraph – Multi-Agent Systems & Building Scalable Agentic Systems (DataCamp) | Cisco C Advanced | Scrum Fundamentals (SCRUMstudy) | Attendance Hashgraph Developer (The Hashgraph Association)
- **Involvement:** IEEE member – Pitch event organizer
- **Languages:** Arabic (native) | English (C1) | French (B2)